

HOME OFFICE NETWORK

Cisco Packet Tracer Lab Report

Computer Networks Project

Topics Covered

VLAN Configuration

DHCP Server Setup

DNS Configuration

Static Routing

Wireless Connectivity

1. Project Overview

This project demonstrates the design and implementation of a Home Office Network in Cisco Packet Tracer. The network separates work and personal devices using VLANs, assigns IP addresses automatically via DHCP, resolves domain names through DNS, and uses static routes to control traffic flow between network segments.

1.1 Objectives

- Separate Work and Personal network traffic using VLANs
- Automatically assign IP addresses using DHCP
- Resolve domain names using a DNS Server
- Control traffic flow using Static Routes
- Provide wireless connectivity using Access Points

1.2 Tools Used

- Cisco Packet Tracer 8.x
- Cisco 2911 Router
- Cisco 2960-24TT Switch
- Generic Server (DNS + DHCP)
- Wireless Access Points
- Laptops with WPC300N wireless modules

2. Network Design

2.1 IP Address Plan

Device / Role	VLAN	IP Address	Subnet Mask	Gateway
Router Fa0/0.10 (Work)	10	192.168.10.1	255.255.255.0	—
Router Fa0/0.20 (Personal)	20	192.168.20.1	255.255.255.0	—
DNS Server	10	192.168.10.2	255.255.255.0	192.168.10.1
DHCP Pool — Work	10	192.168.10.100–150	255.255.255.0	192.168.10.1
DHCP Pool — Personal	20	192.168.20.100–150	255.255.255.0	192.168.20.1
Laptop0 — Work (wireless)	10	DHCP assigned	255.255.255.0	192.168.10.1
Laptop1 — Personal (wireless)	20	DHCP assigned	255.255.255.0	192.168.20.1

Device / Role	VLAN	IP Address	Subnet Mask	Gateway
Access Point 0 (Work)	10	N/A	N/A	N/A
Access Point 1 (Personal)	20	N/A	N/A	N/A

2.2 VLAN Design

VLAN	Name	Network	Purpose
VLAN 10	WORK	192.168.10.0/24	Work devices — Laptop0, DNS Server
VLAN 20	PERSONAL	192.168.20.0/24	Personal devices — Laptop1

2.3 Switch Port Assignment

Switch Port	Connected Device	VLAN	Mode
Fa0/1	Router (Trunk)	10, 20	Trunk
Fa0/2	DNS/DHCP Server	10	Access
Fa0/3	PC0 (Work Test)	10	Access
Fa0/4	PC1 (Personal Test)	20	Access
Fa0/5	Access Point 0 (Work)	10	Access
Fa0/6	Access Point 1 (Personal)	20	Access

3. Configuration Steps

3.1 VLAN Configuration on Switch

VLANs were created and ports were assigned to separate Work and Personal traffic:

```
Switch> enable
Switch# configure terminal
Switch(config)# vlan 10
Switch(config-vlan)# name WORK
Switch(config)# vlan 20
Switch(config-vlan)# name PERSONAL
```

Trunk port configured to carry both VLANs to the Router:

```
Switch(config)# interface FastEthernet0/1
```

```
Switch(config-if)# switchport mode trunk
Switch(config-if)# switchport trunk allowed vlan 10,20
```

3.2 Router Sub-Interface Configuration (Router-on-a-Stick)

Sub-interfaces were created on the router to route traffic between VLANs using a single physical link:

```
Router(config)# interface FastEthernet0/0.10
Router(config-subif)# encapsulation dot1Q 10
Router(config-subif)# ip address 192.168.10.1 255.255.255.0

Router(config)# interface FastEthernet0/0.20
Router(config-subif)# encapsulation dot1Q 20
Router(config-subif)# ip address 192.168.20.1 255.255.255.0
```

3.3 DHCP Configuration

Two DHCP pools were configured on the router — one per VLAN:

```
Router(config)# ip dhcp pool WORK_POOL
Router(dhcp-config)# network 192.168.10.0 255.255.255.0
Router(dhcp-config)# default-router 192.168.10.1
Router(dhcp-config)# dns-server 192.168.10.2

Router(config)# ip dhcp pool PERSONAL_POOL
Router(dhcp-config)# network 192.168.20.0 255.255.255.0
Router(dhcp-config)# default-router 192.168.20.1
Router(dhcp-config)# dns-server 192.168.10.2
```

3.4 DNS Server Configuration

The DNS server was configured at 192.168.10.2 with the following DNS records:

Domain Name	IP Address	Type
www.home.local	192.168.10.2	A Record
work.home.local	192.168.10.1	A Record
router.home.local	192.168.10.1	A Record

3.5 Static Route Configuration

Static routes were added to define explicit traffic paths:

```
Router(config)# ip route 192.168.10.0 255.255.255.0 FastEthernet0/0.10
Router(config)# ip route 192.168.20.0 255.255.255.0 FastEthernet0/0.20
Router(config)# ip route 0.0.0.0 0.0.0.0 Loopback0
```

3.6 Wireless Access Point Configuration

Two Access Points were used instead of WRT300N due to Packet Tracer VLAN limitations:

Access Point	SSID	Security	Password	VLAN
AP0 (Work)	HomeOffice-Work	WPA2-PSK	WorkPass@2024	10
AP1 (Personal)	HomeOffice-Personal	WPA2-PSK	PersonalPass@2024	20

4. Test Results

4.1 Connectivity Test Results

Test	Source	Destination	Expected	Result
Ping — Same VLAN	Laptop0 (Work)	192.168.10.1	Success	PASS
Ping — Cross VLAN	Laptop0 (Work)	192.168.20.1	Success	PASS
Ping — DNS Server	Laptop0 (Work)	192.168.10.2	Success	PASS
Ping — Same VLAN	Laptop1 (Personal)	192.168.20.1	Success	PASS
Ping — Cross VLAN	Laptop1 (Personal)	192.168.10.1	Success	PASS
DNS Resolution	Laptop0	www.home.local	IP returned	PASS
HTTP Web Page	Laptop0	http://www.home.local	Page loads	PASS
nslookup	Laptop0	www.home.local	192.168.10.2	PASS
DHCP — Work Pool	Laptop0 (Work)	DHCP Server	192.168.10.x	PASS
DHCP — Personal Pool	Laptop1 (Personal)	DHCP Server	192.168.20.x	PASS
Wireless — Work	Laptop0	AP0 (HomeOffice-Work)	Connected	PASS
Wireless — Personal	Laptop1	AP1 (HomeOffice-Personal)	Connected	PASS

Test	Source	Destination	Expected	Result
Static Route	Router	show ip route	Routes visible	PASS

4.2 Verification Commands Output

The following commands were used to verify the configuration:

Command	Device	Result
show vlan brief	Switch	VLAN 10 (WORK) and VLAN 20 (PERSONAL) visible
show interfaces trunk	Switch	Fa0/1 and Fa0/5 shown as trunk ports
show ip interface brief	Router	Fa0/0.10 and Fa0/0.20 both up/up
show ip route	Router	Static routes for both VLANs visible
show ip dhcp binding	Router	WORK_POOL and PERSONAL_POOL leases shown
ipconfig	Laptop0	192.168.10.x IP assigned via DHCP
ipconfig	Laptop1	192.168.20.x IP assigned via DHCP

5. Challenges & Solutions

Challenge	Cause	Solution
WRT300N DHCP failed	WRT300N cannot handle VLAN tagged traffic in Packet Tracer	Replaced with two separate Access Points
Tablet has no wireless module	Packet Tracer tablets lack physical module slots	Used Laptop with WPC300N module instead
IP showing 192.168.0.x	WRT300N built-in DHCP was active	Disabled WRT300N DHCP, used router DHCP pools
DNS URL not loading	HTTP service was OFF on server	Enabled HTTP and HTTPS services on server
Cross-VLAN ping failing	Sub-interface encapsulation missing	Re-configured dot1Q encapsulation on sub-interfaces

6. Conclusion

This lab successfully demonstrated the implementation of a Home Office Network with the following key achievements:

- Work and Personal traffic was effectively separated using VLAN 10 and VLAN 20
- Devices automatically received IP addresses from the correct DHCP pool based on their VLAN
- Domain names were successfully resolved using the DNS server at 192.168.10.2
- Static routes enabled controlled inter-VLAN communication through the router
- Wireless connectivity was achieved using two dedicated Access Points for each VLAN
- All ping, DNS, HTTP, and DHCP tests passed successfully

The project provided hands-on experience with enterprise networking concepts including VLANs, inter-VLAN routing (Router-on-a-Stick), DHCP, DNS, and wireless networking in a simulated environment using Cisco Packet Tracer.